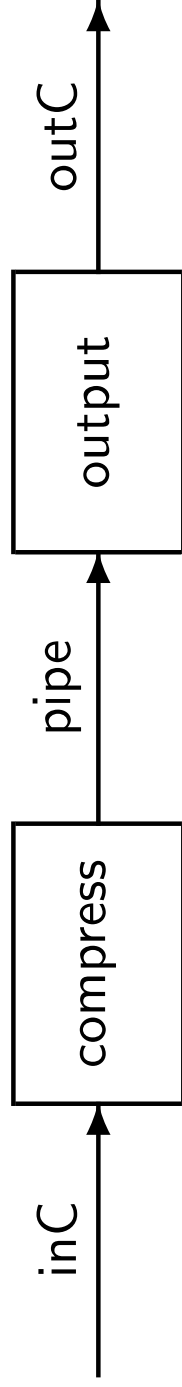


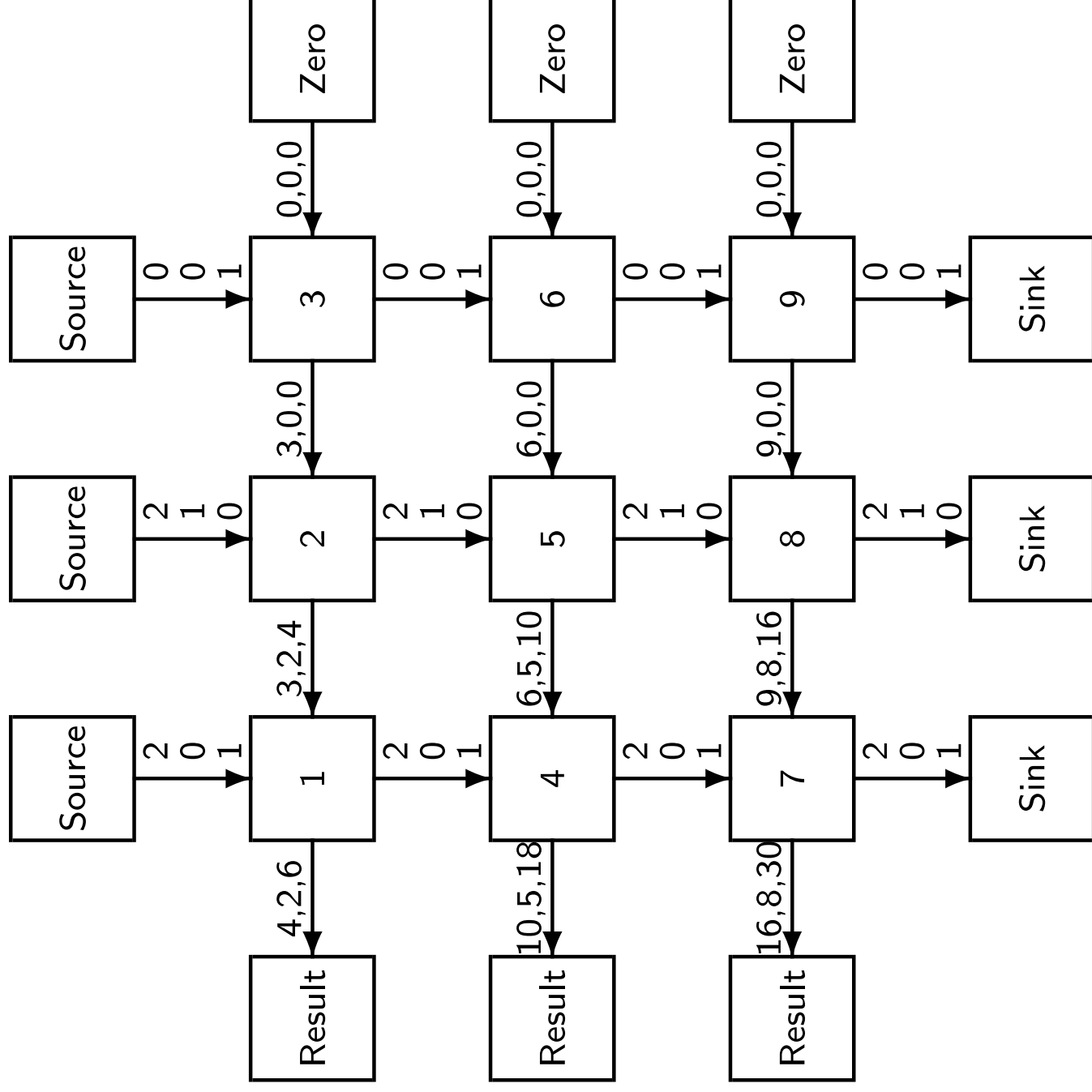
| Algorithm 8.1: Producer-consumer (channels)                    |                                                               |  |
|----------------------------------------------------------------|---------------------------------------------------------------|--|
| channel of integer ch                                          |                                                               |  |
| producer                                                       | consumer                                                      |  |
| integer x<br>loop forever<br>p1:   x ← produce<br>p2:   ch ← x | integer y<br>loop forever<br>q1:   ch ⇒ y<br>q2:   consume(y) |  |

| Algorithm 8.2: Conway's problem                                                                                |                                                                           |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| constant integer MAX $\leftarrow$ 9<br>constant integer K $\leftarrow$ 4<br>channel of integer inC, pipe, outC |                                                                           |
| compress                                                                                                       | output                                                                    |
| char c, previous $\leftarrow$ 0<br>integer n $\leftarrow$ 0<br>inC $\Rightarrow$ previous<br>loop forever      | char c<br>integer m $\leftarrow$ 0                                        |
| p1:    inC $\Rightarrow$ c<br>p2:    if (c = previous) and<br>(n < MAX - 1)<br>n $\leftarrow$ n + 1<br>else    | loop forever<br>q1:    pipe $\Rightarrow$ c<br>q2:    outC $\Leftarrow$ c |
| p3:       n $\leftarrow$ n + 1<br>else                                                                         | q3:    m $\leftarrow$ m + 1                                               |
| p4:       if n > 0                                                                                             | q4:       if m >= K                                                       |
| p5:          pipe $\Leftarrow$ intToChar(n+1)                                                                  | q5:          outC $\Leftarrow$ newline                                    |
| p6:          n $\leftarrow$ 0                                                                                  | q6:          m $\leftarrow$ 0                                             |
| p7:          pipe $\Leftarrow$ previous                                                                        | q7:                                                                       |
| p8:          previous $\leftarrow$ c                                                                           | q8:                                                                       |

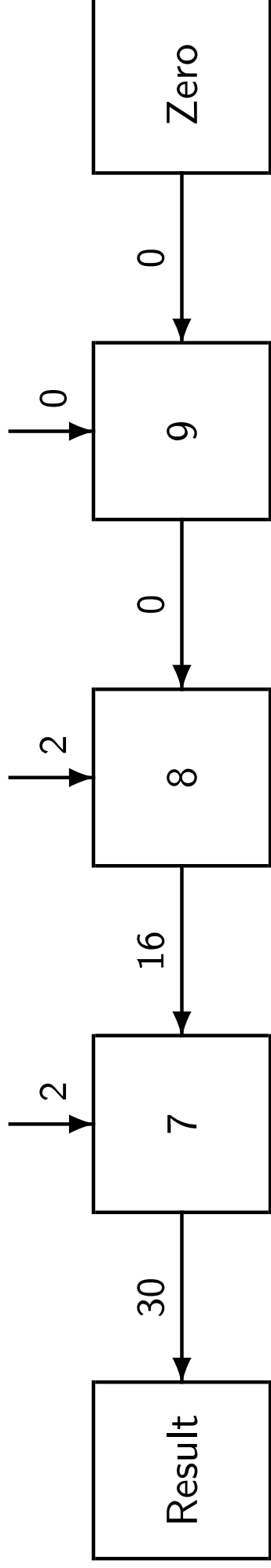
# Conway's Problem



# Process Array for Matrix Multiplication



# Computation of One Element



### Algorithm 8.3: Multiplier process with channels

integer FirstElement  
channel of integer North, East, South, West  
integer Sum, integer SecondElement

loop forever

p1: North  $\Rightarrow$  SecondElement

p2: East  $\Rightarrow$  Sum

p3: Sum  $\leftarrow$  Sum + FirstElement · SecondElement

p4: South  $\Leftarrow$  SecondElement

p5: West  $\Leftarrow$  Sum

### Algorithm 8.4: Multiplier with channels and selective input

integer FirstElement  
channel of integer North, East, South, West  
integer Sum, integer SecondElement

```
loop forever
  either
    p1: North  $\Rightarrow$  SecondElement
    p2: East  $\Rightarrow$  Sum
  or
    p3: East  $\Rightarrow$  Sum
    p4: North  $\Rightarrow$  SecondElement
    p5: South  $\Leftarrow$  SecondElement
    p6: Sum  $\leftarrow$  Sum + FirstElement  $\cdot$  SecondElement
    p7: West  $\Leftarrow$  Sum
```

| Algorithm 8.5: Dining philosophers with channels                                                                                                                                                      |                                                                                                                                 |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--|
| channel of boolean forks[5]                                                                                                                                                                           |                                                                                                                                 |  |
| philosopher i                                                                                                                                                                                         | fork i                                                                                                                          |  |
| boolean dummy<br>loop forever<br>p1: think<br>p2: forks[i] $\Rightarrow$ dummy<br>p3: forks[i+1] $\Rightarrow$ dummy<br>p4: eat<br>p5: forks[i] $\Leftarrow$ true<br>p6: forks[i+1] $\Leftarrow$ true | boolean dummy<br>loop forever<br>q1: forks[i] $\Leftarrow$ true<br>q2: forks[i] $\Rightarrow$ dummy<br>q3:<br>q4:<br>q5:<br>q6: |  |